6. $\vec{r}=\left(3\hat{i}+6\hat{j}-2\hat{k}\right)+s\left(-4\hat{i}-8\hat{j}+8\hat{k}\right)+t\left(3\hat{i}-2\hat{j}\right)s,t\in\mathbb{R}$; $\vec{r}\cdot\left(2\hat{i}+3\hat{j}+4\hat{k}\right)=16$; $2x+3y+4z-16=0$

7. $\vec{r}\cdot\left(3\hat{i}+5\hat{j}-7\hat{k}\right)=6$; $3x+5y-7z-6=0$

பயிற்சி 6.8

1. $\vec{r}\cdot\left(17\hat{i}-47\hat{j}-24\hat{k}\right)=-172$
2. $x+2y-z-4=0$
3. $m=\pm\sqrt{2}$
4. $-2, 2,\ y+z+1=0,\ y-z+1=0$

பயிற்சி 6.9

1. $15x-47y+28z-7=0$
2. $5x-11y+z-17=0$
3. $\sin^{-1}\left(\frac{8}{21}\right)$
4. $\cos^{-1}\left(\frac{2}{3\sqrt{6}}\right)$
5. $2x-3y+5z+11=0, \frac{4}{\sqrt{38}}$
6. $\frac{1}{\sqrt{3}}$ அலகுகள்
7. $(2,2,0),\ \mathrm{Sin}^{-1}\left(\frac{2}{3\sqrt{6}}\right)$
8. $(3,1,-1)$; $\sqrt{14}$ அலகுகள்.

பயிற்சி 6.10

1	2	3	4	5	6	7	8	9	10
(4)	(3)	(1)	(2)	(1)	(3)	(1)	(1)	(1)	(2)
11	12	13	14	15	16	17	18	19	20
(3)	(1)	(2)	(4)	(4)	(2)	(3)	(4)	(2)	(1)
21	22	23	24	25					
(2)	(3)	(4)	(3)	(1)					